

CptS 443/543 Early Data Gathering Report

Eric Furukawa, Harrison Greenlee, Arrin Bevers, Drew Kelly, Verenice Cobian,
Kevin Tran

3/23/2021

Abstract

Our team is developing a clicker game that is intended to appeal to a wide audience, from new players to veterans. In our inquiries, we have determined that upgrades, secondary upgrades, and abilities are often overlooked by new players, which leads to a poor gameplay experience. We have also found that new players will lose interest if not “hooked” by an exciting burst of progress. However, our inquiry has also shown that veteran players expect long-term progression and complex gameplay interactions. Based on these observations, it is clear that a good clicker game must be easy to learn while still providing sufficient depth to satisfy veteran players.

Design project focus and research questions

a. The problem the software aims to address

Idle games such as Cookie Clicker are highly entertaining when executed well, but often are held back by confusing and unappealing UI and related game mechanics. Our team aims to create an idle game that uses crisp and functional UI to pull in more players and keep them playing after a short learning period.

b. Proposed software solution description

Resource Gathering

Sometimes resource gathering may be boring to the user without proper motivation or interesting things to do. Although not necessarily related to UI, the game may feature mechanics such as auto battling and branching upgrade paths to keep the player more interested. This does relate to UI later however as these will need nice indicators in the UI as well.

Resource Gain Indicators

Resource gain indicators will be easy to see as done in some of the current clicker games.

Resource gain indicators will always be visible at a central location.

Upgrade/Progress Displays

Current games have the problem of users not noticing what the upgrades are or what they do until they actually buy them. We want to have a way to display them in a way that users are almost guaranteed to know what they are and they do.

Adding narrative/story elements

Most clicker games don't have commonly occurring story progressions so having our game have frequent story updates will keep the player invested and interested both within one play session and over the span of the entire playtime.

Uniqueness in UI

Menus that unfold or slide open allow for a minimalist UI interface that can also allow for depth in choices.

Interesting button press and upgrade animations will be added to give the game a unique flair to the game feel.

Ease of Use Changes

When the game first starts, there will be indicators that will ensure the user does not miss any critical UI elements, and/or a tutorial to introduce players to the essential features and walk them through the expected progression.

c. Related products that address the problem

The games we're trying to "fix" are idle/clicker/incremental games such as [Cookie Clicker](#), [A Dark Room](#), [Clicker Heroes](#), and [Realm Grinder](#). These are games where the user tries to gain resources with a combination of waiting (idling) and doing basic tasks (clicking) and using those resources to purchase upgrades which make said actions more rewarding, earning more resources. The user is meant to figure out how to achieve this in the most efficient way possible. As noted, due to issues with the UI, users may not be able to learn how to do this.

d. Proposed software's prospective users

Our target demographic is primarily the casual game audience who want to be able to occasionally pick up and play a simple game in their spare time. This demographic is extremely wide and can include all age ranges excluding extremely young children or any gender. Most users will have at least some passing interest in video games and in general like optimization in games.

There are two major secondary targets to keep in mind as well. The "hardcore" video game audience may also enjoy playing the game if it presents itself to be capable of mechanical depth after the initial easy learning curve. Additionally, the overlapping group of users who will be interested due to the zany nature of the game will be more convinced to play by introducing more in-depth aspects of the game. The interviewees reflect the varying experience levels of users expected to play our game.

e. The key research questions that the contextual inquiry set to answer

RQ1 How much does the theme of an idle game determine the fun the user has and which themes might be best (In specific, would steampunk work)?

RQ2 How can the feedback of clicking the button be made more satisfying?

RQ3 What type of menu layout and design will both not get in the way of the player and show satisfying progress?

RQ4 What are some menu options that grab too much or too little attention from the user and why?

RQ5 What do some other idle games' menus do uniquely best/worst and can we apply the best and avoid the worst in our game?

RQ6 Can such a game work on both mobile, executable and web browser or do certain games restrict this? If so, what?

RQ7 What are features missing from clicker games that could be added?

Participants

Participant 1 is a fourth-year undergraduate computer science major at WSU. Notably, they have not taken the CptS 443 course as of this inquiry. The participant is a hard working student who likes playing video games and has played some idle games such as Cookie Clicker before, albeit quite some time ago. Although they enjoy clicker games, it is not their first choice when it comes to playing video games.

Participant 2 is a middle-aged adult with a high school diploma and no college experience. This individual's total video-game related experience is Frogger and Pong. Obviously, this means they had no prior experience with Cookie Clicker as well. Their typical leisure time involves cooking and late-night TV shows.

Participant 3 is a fourth-year undergraduate English major at WSU. They have not taken CptS 443 as of this inquiry. The participant likes playing other games but they have no experience with clicker games.

Participant 4 is not in college. They spend a significant amount of time every day playing video games, and have a lot of experience with clicker games.

Participant 5 is an adult gamer who has not played idle/clicker games before. They have quite a lot of experience playing other kinds of games, and are interested in game design, having used tools like GameMaker to develop small games of their own.

Participant 6 is a first-year graduate student at Arizona State University. They are more of a trend gamer and use gaming as a means to socialize with friends. They have no prior experience with idle/clicker games.

Contextual Inquiry Sessions

Interview Script

Before Starting

- Have you played this kind of game before?

If so:

1. What influences which games in this genre you choose to play?
2. What mood are you generally in when you play games like these?
3. How do games like this achieve mechanical depth? What makes them interesting?
4. Do you prefer random luck-based progression or steady time-based progression?

If not:

1. What do you expect this game to be about?
2. What is your opinion of this genre?

During Play

- What made you do this action (What were you thinking)?
- How did you know to do [action] to achieve [result]?
- What is grabbing most of your attention?
- What stands out as the most confusing? (*I think we should ask both at the end as an overall what was the most confusing thing, but also as they play, what is confusing or frustrating right now).
- Is there anything about the interface that annoys you?

After Play

- (If this is the first game like this that they have played) - Has your opinion of the genre changed at all, or do you still feel the same way about it?

- Would you return to this website to play again? Why?
- Are there any gameplay features that you wish were in the game?
- If you could change one thing about the interface, what would it be?
- What did you like the most?
- What did you like the least?
- How intuitive was the interface? Were you confused by anything in particular?
- Would you play a game like this if it was a mobile app? What about if it was a windows executable? (If the game isn't on a website, ask about that too).
- If the cookies were replaced with something else, would it influence your enjoyment?
- If the game had a steampunk theme, what features would you expect? Any cool ideas?

Interview 1

Interview happened in a group voice chat on Discord. The participant was asked to sign the consent form and then to open the game, A Dark Room. The game automatically started so the participant was asked to close the game and then answer the set of “Before Playing” questions. The user expressed general interest in games but perhaps a little bit of skepticism on the minimalistic UI idle game they were to be playing.

The participant then proceeded to open the game and start playing it. The participant was very quick to learn the beginning game loop in order to start making progress. Despite the game not telling the user what anything did in the grand scheme of things, they continued to purchase upgrades with some logic that “just made sense” to them. Some intermittent questions such as “What is grabbing your attention most?” and “How did you know to do this?” were asked. Many questions naturally stemmed from what I was wondering about the game as well. After more questions, with “Why did you do this?” probably being the more occurrent one (although many different ones were asked), the gameplay portion of the interview ended.

The participant was then asked the “After Playing” questions.

Interview 2

This interview took place in a physical location at a dedicated computer terminal. The participant was given the signed consent form after sitting down into the chair in front of the keyboard. The designer read aloud the consent form and requested verbal understanding of the requirements outlined in the document before proceeding.

Once the game (Cookie Clicker) began, the designer sat behind the user and passively observed while they played the game. Aside from occasionally asking questions about the user's actions, the majority of the questions came during and after the game itself. This included some questions about using basic computer peripherals since the user is not familiar with using computer hardware for extended periods of time (exceeding a few minutes).

Throughout the entirety of the game and after, the user didn't have significant input on the game itself, which was somewhat disappointing. Some aspects of the game were more confusing for them, mostly the store upgrades, but they said they remained invested because of the cookie imagery. I think this was the most interesting finding, especially since this user is not a “gamer”.

Interview 3

The interview took place over Zoom. The participant was asked to sign the consent form and then asked if they had played clicker games before. After responding, they were told to open Clicker Heroes via Steam.

Once the game began, the interviewer observed the interviewee's gameplay experience, asking questions periodically throughout such as "why did you do that?" and "what are you focusing on right now?". Additionally, the user was asked UI questions from the script. Although the user missed critical UI elements, they were not informed of this until after the game.

After the in-game interview, they were asked questions from the post-game script. The subject was allowed to continue playing the game as they responded to these questions.

Interview 4

This interview took place over Zoom. The participant was asked to sign the consent form and then asked if they had played clicker games before. After affirming that they have, they were told to open their favorite clicker game. They opened Realm Grinder via Steam and began to explain the mechanics of the game as they played.

After their initial explanation of the general gameplay and mechanics, they were asked questions as they played such as why did you do that?" and "what are you focusing on right now?". Additionally, the user was asked UI questions from the script.

After the in-game interview, they were asked questions from the post-game script. The subject was allowed to continue playing the game as they responded to these questions.

Interview 5

The interview took place over Zoom. The interviewee was directed to load the Cookie Clicker game in a web browser, though not to begin playing until the recording started. The interviewee was given a brief description of the process:

They would be asked a few preliminary questions. Then they would play the game and were encouraged to describe their process and experience as they were playing. The interviewer may ask them questions such as "why did you choose to do that?" during play. They would play for about 10 to 15 minutes before moving on to the follow-up questions.

The interview stayed mainly on script, the interviewer attempted not to direct the gameplay of the interviewee, but did ask, for example, after the interviewee did not interact with a time sensitive gameplay element whether they had seen it and chose not to interact with it, or whether they had not seen it. The interviewee continued to play while follow-up questions were asked.

Interview 6

The interview took place in a physical location at a private residence. The participant was asked to read the consent form. After reading it, they then signed a blank sheet of paper which I later scanned and added to the consent document. I then gave some background on the goal of the team project and provided an outline of how the interview will go.

I asked the participant the questions designed to be asked before playing the clicker game. Overall, they had no prior experience with clicker games. I then opened up Cookie Clicker and had them start playing. At first, the participant inspected the page, presumably looking for instructions on what to do. After not finding any, they clicked the cookie and discovered that each click earned them points. After about a minute, I began asking the questions meant for during gameplay. The participant was not impressed by the game. The participant stopped playing and answered the "After Play" questions.

Common Tasks and Themes

- New users tend to miss the UI elements for secondary upgrades, which causes serious issues with the pacing and progression of the game.
- New users tend to make purchases in a sub-optimal order, which causes serious issues with the pacing and progression of the game.
- New users tend to question the entertainment value of the game initially, especially if they don't notice the secondary upgrade functionality.
- All users tend to start gameplay by clicking the resource gathering button repeatedly.
- In the early game, after acquiring enough resources to make a purchase, users generally spent them immediately.

Unique features of individual CI sessions

Interview 1

For the first interview, the recording settings were not checked beforehand and led to the interviewer side audio being unrecorded. It was also difficult to ask questions without taking the user "out of the zone". Switching between the recording software, the shared screen of the gameplay and the questions document was at times difficult to juggle. It was fortunate that the participant was interested throughout the entire process as it made interviewing feel effective for the interviewer and not torturous to the participant.

The participant increasingly became interested in the mechanics of the game as it steadily ramped up the complexity of the tasks/systems. They explained that items unlocked later and lower on the list as goals to get and some being more effective than others. At this point the participant started to remark that the game was very different from what they expected and continued to play for longer than expected.

The user was definitely surprised with the depth of not only the mechanics, but the world building the game had set forth as they had not expected that from an idle game. The participant expressed mixed feelings about the minimalistic approach the game took to UI. The minimalist look was unique to the game and added to the feel and allure of it. However, the participant wanted to see at least some visuals like maybe the flame of the fire from the game. Another conflict was them wanting to see what upgrades did exactly, but "understood what they were going for" leaving those details ambiguous. The participant also said a similar game would be best on a web browser. The setting wasn't particularly important to the game for them and thought a steampunk setting could work just as well. They suggested starting out in a wasteland or garbage dump, gathering fuel and making interesting machinery. As a final tip, they said to focus on the "ooh shiny" kind of feeling of such a game and to have either a good narrative or good visuals/theming to carry the game (ideally both).

Interview 3

Interviewee greatly enjoyed killing a mouse enemy who makes a funny squeak sound when they die. Interviewee stated that the genre was boring after the gameplay session was complete.

Interview 4

Interviewee seemed to have a lot of experience with clicker games, and was able to navigate his game's complex user interface with relative ease. Interviewee strongly encouraged us to include a "buy multiple" functionality in our clicker game.

Interview 5

Interviewee reflected on gameplay elements from a somewhat ironic and philosophical perspective, eg when informed that a player could return to the game and their progress would be preserved the interviewee remarked on “Cookie Clicker” being a pun. Interviewee seemed more entertained by musing about the possibilities for improving the game than playing the game itself.

Synthesis of findings Requirements

Table 1. Functional Requirements and Associated Usability Targets

Functional Requirement	Associated Usability Target(s)	Empirical Source/Rationale
Users must be able to gain resources by clicking the primary button.	Users must be able to click the resource button every time their upgrades allow them to (variable time).	Interviewed users indicated that they expect this functionality from a game in the genre.
Users must be able to spend resources on an in-game shop on items by navigating to the upgrade.	Users must be able to spend resources on an item from the item menu within 5 seconds.	Interview subjects frequently expressed skepticism about the entertainment value of the game at the very beginning, but their interest in the game increased noticeably when they interacted
Users must be able to view their current status (current resources and items).	Users should always be able to view their current resource. Users should be able to see their current items within 3 seconds.	Interviewed users stated that they are constantly focusing on their resources while playing.
Users must be able to resume their progress.	Users should be able to resume their progress within 10 seconds of starting the program.	Users stated that their goal was to accumulate as much money as possible. If the game resets after every session, this goal is not achievable.
Users must be able to swap between the shop and button/progress views.	Users should be able to swap between the shop and	Users often missed secondary upgrades. By allocating more space to a dedicated

	button/progress view within 2 seconds.	shop view, we hope to increase the usability of the shop UI.
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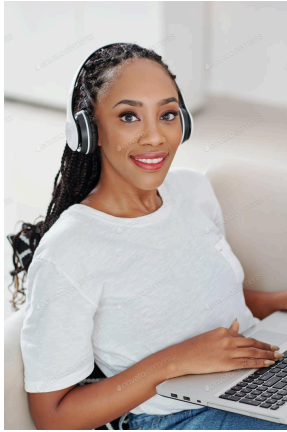
Table 2. User Experience Requirements

User Experience Requirement	Empirical Source/Rationale
On a scale from 1 to 10, users must rate the app a 7 in terms of entertainment value.	The core goal of a game in this genre is to entertain the user. If the game was not fulfilling this requirement, interviewed users expressed a desire to stop playing.
On a scale from 1 to 10, users must rate the app a 9 in terms of ease of use.	Several of our interviewees were confused by UI elements in other clickers.

Personas

 "I used to play a lot of games back in high school and college but I just don't have time anymore" Goals - Family and Friends first - Just relaxing a bit - Playing with friends	Basic Info Software Engineer Casual mobile, PC and console player Who Are They? Ashley is a software engineer for a tech startup. They recently graduated with a Bachelor's and dedicate most of their outside time to their social + family life and playing soccer. They only play games on the subway on their way to and from work and 2-3 hours each week on their computer or Playstation. They will try to play some with their friends online when possible. Although they used to like intensive games, they now prefer easy to learn games without time commitments.	Ashley David Game Profile Can pick up most less complicated games pretty fast as they know most of the basic control setups. They even have a bit of experience in how games are made when they made one in college for a class. They vaguely remember what Cookie Clicker is but didn't play it themselves and generally don't play browser games. They also don't really look for new games to play as the ones they do play are good enough unless if their friends tell them of a new one to try. Games They Play - Angry Birds - Bloons TD - Minecraft - Call of Duty Doesn't Like - Hard to learn mechanics. - Obnoxious amounts of visuals. - Needing to "keep up" with game updates/strategy.
 "I have been playing Cookie Clicker since it was released in 2013." Goals - Finding entertaining games to play during her breaks - Getting as many cookies as possible	Basic Info Bus Driver PC and console player, with a focus on clicker games Who Are They? Mindy is a school bus driver for Northbridge elementary school. She was first introduced to clicker games when she was waiting for a long flight. She likes playing clickers while waiting for students to get out of school. When she gets home, she plays other games, but usually has at least a couple clicker games running in the background.	Mindy Lucero Game Profile A dedicated clicker-game enthusiast. Has played multiple clicker games and is familiar with the mechanics and conventions of the genre. Although they play other games they tend to have a couple of clickers running in the background. Games They Play - Cookie Clicker - A Dark Room - Realm Grinder - Clicker Heroes - Call of Duty - Minecraft Doesn't Like - Games with overly simplistic mechanics. - Games that lack depth and complexity.
 "This seems like a waste of time." Goals - Fishing - Reading - Other entertainment	Basic Info Retired Chemist Dedicated Fisherman Who Are They? Intelligent, but not tech savvy. Frank has a PhD in Chemistry, but trouble navigating the internet. He uses an old laptop to connect to Facebook to stay in touch with his family. His kids try to get him to play computer games, but they are too confusing, so he doesn't find them to be entertaining most of the time.	Frank Douglas Game Profile Enjoys playing poker with his friends every Friday. Also likes playing board games with family, but doesn't usually play computer games because he doesn't have a lot of patience for complex or obscure game mechanics. He's skeptical of video games, though open to being convinced, if you can capture his attention before he gets bored. Games They Play - Monopoly - Poker - Settlers of Catan - Risk Doesn't Like - Whatever a "Pikachu" is. - Using computers. - Downloading software.

Scenarios

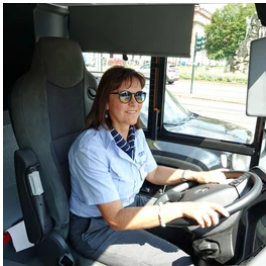


What matters to Ashley when playing a clicker game?

Ashley used to enjoy playing computer games in high school and college but since starting her job as a software engineer she finds she doesn't have the time to dedicate to gaming that she used to. She misses the stories and the sense of accomplishment, the excitement of "leveling up," but finds that most games demand a prohibitive time investment.

She's curious about clicker games because they seem to have some of the gameplay elements she's interested in, but don't require that she spend hours playing to advance.

She's looking for something that's easy to pick up and put down, something that has some kind of interesting story or other creative dimension, and something that she can make progress at without having to devote time to it.



What matters to Mindy when playing a clicker game?

Mindy has played a number of other clicker games, and although she enjoys the genre she's looking for something that will stand out. She's got a few favorites, but she also thinks that they could be improved. She'd love to see a clicker game that delivered on the things that she's liked from other games, but also includes something new, that makes it worth playing over (or in addition to) established favorites.

Mindy is looking for something funny or novel, or a new twist on the genre that will freshen the experience up a bit.



What matters to Frank when playing a clicker game

Frank wants something to entertain him while the lake is frozen over and he can't fish. His kids have encouraged him to download a clicker game to pass the time, and he is excited by the prospect of discussing the game with them. He is unsure about playing a video game, citing bad experiences in the past with poorly designed video games, but eventually he agrees to give it a shot.

Frank hasn't played enough clicker games to fully understand their conceptual model. He knows they involve clicking, in some capacity, but he isn't sure why clicking something a million times would be entertaining. He also has no experience with upgrade systems and he wants to avoid complicated UI elements, so unless he is directed to interact with them, he will avoid them.

Sudden spikes of early progress are likely to intrigue Frank and encourage him to continue playing.

Appendix A: Informed Consent

Informed Consent Agreement to Participate In Contextual Inquiry

[Arrin Bevers, Verenice Cobian, Eric Furukawa, Harrison Greenlee, Denise Michelle Kelly, Kevin Tran]
School of Electrical Engineering and Computer Science
Washington State University

Description of Study: I understand that I, Timmy Huynh, have been asked to participate in a contextual inquiry to inform the design of a new software application being created as part of the above persons' (henceforth, "the designers") course project for CptS 443/543 at Washington State University. My participation in this activity will help the designers to better understand the needs of prospective users of the software. I have been asked to spend about 30 minutes participating in this test. This will involve my engaging in user interface testing while the designers observe, ask questions, and take notes.

The designers will record the session on videotape. My name will not be on the videotape. When the designers describe their work to other people in class (which may entail showing segments of the videotape), they will not use my name.

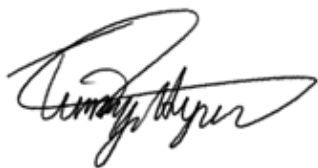
Risks and Benefits Expected: The contextual inquiry will not do me any harm. It is not expected to help me directly. The results may help inform the design of the designers' software.

Confidentiality: I understand that any information about me that is obtained from this contextual inquiry, including what I say, will be confidential. My real name will be kept in a locked file and only the researchers will have access to it. Only my code name will be associated with data collected on me. Reports and presentations involving those data will not use my real name and will not present other data that could be used to identify me. Any recordings made within this contextual inquiry will be destroyed within two years.

Right to Refuse or End Participation: I understand that I may refuse to participate in this study or stop participating at any time.

Certification: I certify that I have read and that I understand the foregoing, that I have been given satisfactory answers to my inquiries concerning this contextual inquiry, and that I have been advised that I am free to withdraw my consent and to discontinue participation in the project or activity at any time.

I herewith give my consent to participate in this activity with the understanding that such consent does not waive any of my legal rights, nor does it release the researchers or any agent thereof from liability for negligence. I understand that I shall remain anonymous in all written and verbal reports of this study. If I am recorded, I agree to allow the designers to present to their instructor and classmates excerpts of any recordings taken during the study for educational purposes. I understand that I may request a copy of this form to keep.



3/17/2021

Signature of individual participant

Date

(If you cannot obtain satisfactory answers to your questions or have comments or complaints about your treatment in this activity, please contact Professor Christopher Hundhausen, Washington State University, 509-335-4590 or hundhaus@wsu.edu.)

Informed Consent Agreement to Participate In

Contextual Inquiry

Arrin Bevers, Verenice Cobian, Eric Furukawa, Harrison Greenlee, Denise Michelle Kelly, Kevin Tran
School of Electrical Engineering and Computer Science
Washington State University

Description of Study: I understand that I, Jeanie Bevers have been asked to participate in a contextual inquiry to inform the design of a new software application being created as part of the above persons' (henceforth, "the designers") course project for CptS 443/543 at Washington State University. My participation in this activity will help the designers to better understand the needs of prospective users of the software. I have been asked to spend about 30 minutes participating in this test. This will involve my engaging in user interface testing while the designers observe, ask questions, and take notes.

Risks and Benefits Expected: The contextual inquiry will not do me any harm. It is not expected to help me directly. The results may help inform the design of the designers' software.

Confidentiality: I understand that any information about me that is obtained from this contextual inquiry, including what I say, will be confidential. My real name will be kept in a locked file and only the researchers will have access to it. Only my code name will be associated with data collected on me. Reports and presentations involving those data will not use my real name and will not present other data that could be used to identify me. Any recordings made within this contextual inquiry will be destroyed within two years.

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3/16/2021

Signature of individual participant

Date

(If you cannot obtain satisfactory answers to your questions or have comments or complaints about your treatment in this activity, please contact Professor Christopher Hundhausen, Washington State University, 509-335-4590 or hundhaus@wsu.edu.)

Informed Consent Agreement to Participate In

Contextual Inquiry

~~Arrin Bevers, Verenice Cobian~~, Eric Furukawa, Harrison Greenlee, Denise Michelle Kelly, Kevin Tran
School of Electrical Engineering and Computer Science
Washington State University

Description of Study: I understand that I, Grace LaPierre, have been asked to participate in a contextual inquiry to inform the design of a new software application being created as part of the above persons' (henceforth, "the designers") course project for CptS 443/543 at Washington State University. My participation in this activity will help the designers to better understand the needs of prospective users of the software. I have been asked to spend about 40 minutes participating in this test. This will involve me engaging in a clicker game while the designers observe, ask questions, and take notes.

The designers will record the session. My name will not be on the videotape. When the designers describe their work to other people in class (which may entail showing segments of the videotape), they will not use my name.

The designers will record the session on audiotape. My name will not be on the audiotape. When the researchers describe their work to other people in class (which may entail playing segments of the audiotape), they will not use my name.

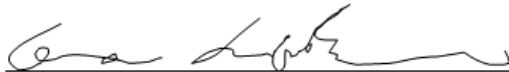
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Right to Refuse or End Participation: I understand that I may refuse to participate in this study or stop participating at any time.

Certification: I certify that I have read and that I understand the foregoing, that I have been given satisfactory answers to my inquiries concerning this contextual inquiry, and that I have been advised that I am free to withdraw my consent and to discontinue participation in the project or activity at any time.

I herewith give my consent to participate in this activity with the understanding that such consent does not waive any of my legal rights, nor does it release the researchers or any agent thereof from liability for negligence. I understand that I shall remain anonymous in all written and verbal reports of this study. If I am recorded, I agree to allow the designers to present to their instructor and classmates excerpts of any recordings taken during the study for educational purposes. I understand that I may request a copy of this form to keep.



Mar 20, 2021

Signature of individual participant

Date

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Informed Consent Agreement to Participate In

Contextual Inquiry

~~Arris Bevers, Verence Cobian, Eric Furukawa, Harrison Greenlee, Denise Michelle Kelly, Kevin Tran~~
School of Electrical Engineering and Computer Science
Washington State University

Description of Study: I understand that I, Daniel Russell have been asked to participate in a contextual inquiry to inform the design of a new software application being created as part of the above persons' (henceforth, "the designers") course project for CptS 443/543 at Washington State University. My participation in this activity will help the designers to better understand the needs of prospective users of the software. I have been asked to spend about 40 minutes participating in this test. This will involve me engaging in a clicker game while the designers observe, ask questions, and take notes.

The designers will record the session. My name will not be on the videotape. When the designers describe their work to other people in class (which may entail showing segments of the videotape), they will not use my name.

The designers will record the session on audiotape. My name will not be on the audiotape. When the researchers describe their work to other people in class (which may entail playing segments of the audiotape), they will not use my name.

Risks and Benefits Expected: The contextual inquiry will not do me any harm. It is not expected to help me directly. The results may help inform the design of the designers' software.

Confidentiality: I understand that any information about me that is obtained from this contextual inquiry, including what I say, will be confidential. My real name will be kept in a locked file and only the researchers will have access to it. Only my code name will be associated with data collected on me. Reports and presentations involving those data will not use my real name and will not present other data that could be used to identify me. Any recordings made within this contextual inquiry will be destroyed within two years.

Right to Refuse or End Participation: I understand that I may refuse to participate in this study or stop participating at any time.

Certification: I certify that I have read and that I understand the foregoing, that I have been given satisfactory answers to my inquiries concerning this contextual inquiry, and that I have been advised that I am free to withdraw my consent and to discontinue participation in the project or activity at any time.

I herewith give my consent to participate in this activity with the understanding that such consent does not waive any of my legal rights, nor does it release the researchers or any agent thereof from liability for negligence. I understand that I shall remain anonymous in all written and verbal reports of this study. If I am recorded, I agree to allow the designers to present to their instructor and classmates excerpts of any recordings taken during the study for educational purposes. I understand that I may request a copy of this form to keep.

Daniel Russell

Mar 20, 2021

Signature of individual participant

Date

(If you cannot obtain satisfactory answers to your questions or have comments or complaints about your treatment in this activity, please contact Professor Christopher Hundhausen, Washington State University, 509-335-4590 or hundhaus@wsu.edu.)

Informed Consent Agreement to Participate In

Contextual Inquiry

Eric Furukawa, Harrison Greenlee, Arrin Bevers, Drew Kelly, Verenice Cobian, Kevin Tran
School of Electrical Engineering and Computer Science
Washington State University

Description of Study: I understand that I, Nick Thompson have been asked to participate in a contextual inquiry to inform the design of a new software application being created as part of the above persons' (henceforth, "the designers") course project for CptS 443/543 at Washington State University. My participation in this activity will help the designers to better understand the needs of prospective users of the software. I have been asked to spend about 20 minutes participating in this test. This will involve my engaging in an online "clicker" game while the designers observe, ask questions, and take notes.

The designers will record the session via zoom. My name will not be on the recording. When the designers describe their work to other people in class (which may entail showing segments of the video), they will not use my name.

Risks and Benefits Expected: The contextual inquiry will not do me any harm. It is not expected to help me directly. The results may help inform the design of the designers' software.

Confidentiality: I understand that any information about me that is obtained from this contextual inquiry, including what I say, will be confidential. My real name will be kept in a locked file and only the researchers will have access to it. Only my code name will be associated with data collected on me. Reports and presentations involving those data will not use my real name and will not present other data that could be used to identify me. Any recordings made within this contextual inquiry will be destroyed within two years.

Right to Refuse or End Participation: I understand that I may refuse to participate in this study or stop participating at any time.

Certification: I certify that I have read and that I understand the foregoing, that I have been given satisfactory answers to my inquiries concerning this contextual inquiry, and that I have been advised that I am free to withdraw my consent and to discontinue participation in the project or activity at any time.

I herewith give my consent to participate in this activity with the understanding that such consent does not waive any of my legal rights, nor does it release the researchers or any agent thereof from liability for negligence. I understand that I shall remain anonymous in all written and verbal reports of this study. If I am recorded, I agree to allow the designers to present to their instructor and classmates excerpts of any recordings taken during the study for educational purposes. I understand that I may request a copy of this form to keep.

Nick Thompson
Digitally signed by Nick Thompson
DN: cn=Nick Thompson, c=US,
email=pompadourslim@gmail.com
Date: 2021.03.21 13:56:47 -07'00'

Signature of individual participant

Date

(If you cannot obtain satisfactory answers to your questions or have comments or complaints about your treatment in this activity, please contact Professor Christopher Hundhausen, Washington State University, 509-335-4590 or hundhaus@wsu.edu.)

Detail of digital signature:

Nick Thompson
Digitally signed by Nick Thompson
DN: cn=Nick Thompson, c=US,
email=pompadourslim@gmail.com
Date: 2021.03.21 13:56:47 -07'00'

Signature of individual participant

Informed Consent Agreement to Participate In

Contextual Inquiry

Arrin Bevers, Verence Cobian, Eric Furukawa, Harrison Greenlee, Denise Michelle Kelly, Kevin Tran
School of Electrical Engineering and Computer Science
Washington State University

Description of Study: I understand that I, _____ Gerson Leonel _____ have been asked to participate in a contextual inquiry to inform the design of a new software application being created as part of the above persons' (henceforth, "the designers") course project for CptS 443/543 at Washington State University. My participation in this activity will help the designers to better understand the needs of prospective users of the software. I have been asked to spend about 30 minutes participating in this test. This will involve my engaging in user interface testing while the designers observe, ask questions, and take notes.

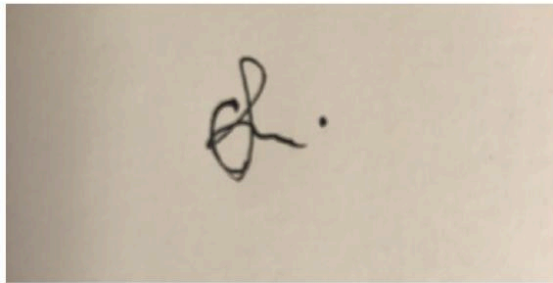
Risks and Benefits Expected: The contextual inquiry will not do me any harm. It is not expected to help me directly. The results may help inform the design of the designers' software.

Confidentiality: I understand that any information about me that is obtained from this contextual inquiry, including what I say, will be confidential. My real name will be kept in a locked file and only the researchers will have access to it. Only my code name will be associated with data collected on me. Reports and presentations involving those data will not use my real name and will not present other data that could be used to identify me. Any recordings made within this contextual inquiry will be destroyed within two years.

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A handwritten signature in black ink on a light-colored background. The signature is stylized, starting with a large 'C' and ending with a period.

3/17/2021

Signature of individual participant

Date

(If you cannot obtain satisfactory answers to your questions or have comments or complaints about your treatment in this activity, please contact Professor Christopher Hundhausen, Washington State University, 509-335-4590 or hundhaus@wsu.edu.)

Appendix B: Raw Data
Participant 1

Video: <https://www.youtube.com/watch?v=1IL4INxCqP4> (Video is Unlisted (only those with links can view), any sections with names are cut out)

Game: A Dark Room

Questions

Before Starting

Have you played this kind of game before?

If so:

1. What influences which games in this genre you choose to play?

The most important aspect is a theme or some nice visuals. Visually appealing.

2. What mood are you generally in when you play games like these?

More relaxed, multitasking, while watching something. Keep hands busy.

3. How do games like this achieve mechanical depth? What makes them interesting?

Mechanically deep? Mostly clickers aren't. Auto clickers and stuff like that basically. Manage money and max efficiency.

4. Do you prefer random luck-based progression or steady time-based progression?

Steady time based progression. Don't mind luck to keep things refreshing.

If not (had them answer these anyways):

1. What do you expect this game to be about?

Keeping a fire going, next to a forest. Next to a room. Keeping up with wood. More complexity to follow?

2. What is your opinion of this genre?

It was a surprising game for the genre. So many games. Played cookie clicker before. Dumb game to play but it's grown surprising amounts. It has its niche. A distraction either good or bad.

During Play (In video)

What made you do this action (What were you thinking)?

How did you know to do [action] to achieve [result]?

What is grabbing most of your attention?

What stands out as the most confusing? (*I think we should ask both at the end as an overall what was the most confusing thing, but also as they play, what is confusing or frustrating right now).

Is there anything about the interface that annoys you?

After Play

(If this is the first game like this that they have played) - Has your opinion of the genre changed at all, or do you still feel the same way about it?

In general, feel the same, but it was unique for a clicker, had a story. Really interesting, how far would it build up? Usually it's just one story.

Would you return to this website to play again? Why?

If really bored, probably. Really interesting because of simple UI, alluring. Seems easy but ramps up super quick.

Are there any gameplay features that you wish were in the game?

More descriptors. What is going on, but what was this going for? Maybe more stats when hovering over.

If you could change one thing about the interface, what would it be?

Strictly text UI, but would be nice to see something on the story. Maybe just the fire. Want to see the progression!

What did you like the most?

Story/progression in a sense. Seems to ramp up really fast after building a few things.

What did you like the least?

Speed of gathering supplies felt a bit long. Time in between long. Faster way to speed these up early on. Being able to increase those things to become faster big part of the game.

How intuitive was the interface? Were you confused by anything in particular?

Not about the interface itself, just the gameplay. What to prioritize?

Would you play a game like this if it was a mobile app? What about if it was a windows executable? (If the game isn't on a website, ask about that too).

Doesn't need to be on steam. Is simple so it should be on browser. Would be a bit odd looking on mobile.

If the cookies were replaced with something else, would it influence your enjoyment?

Would probably still work. Forest wasn't the particularly interesting part. The progression is the important part and the presentation too.

If the game had a steampunk theme, what features would you expect? Any cool ideas?

If it had to be steampunk... having character/theme set in wasteland. In a garbage dump. Would work, gathering fuel and interesting to see machinery and such.

On the unique minimalist aspect: Nostalgic of early 90s stuff, text based games. Could understand the theme it was going for. Liked the visual of loading the wood and stuff. Would like to see the fire since it was so integral to the story. But the text gave plenty of information.

General tips: Have an interesting story to build around or have good visuals. The two things to keep the player invested. The "ooh shiny" aspect. Very surprised by the game.

Participant 2

Game: Cookie Clicker

Questions

Before Starting

Have you played this kind of game before?

“No”

During Play

What made you do this action?

“Since it’s called Cookie Clicker, I clicked on the cookies”

What is grabbing most of your attention?

“The numbers and the cookie pictures”

What stands out as the most confusing? (*I think we should ask both at the end as an overall what was the most confusing thing, but also as they play, what is confusing or frustrating right now)

“The upgrades don’t make any sense to me”

After Play

How intuitive was the interface? How did you know what to do?

“The store is confusing and I don’t understand what the upgrades do”

What did you like the most?

“Watching the cookies increase in the background”

What was the most frustrating?

“What is the goal?”

What was the most confusing?

“What are the cookies being used for?”

Would you return to this website to play again? Why?

“Maybe, it seems kind of pointless”

Is there anything in specific you would like to see in addition to the current functionality?

“More ways to interact with the cookies outside of clicking”

If the cookies were replaced with ____, would it influence your enjoyment?

“Replacing the cookies with piggies!”

Participant 3

Game: Clicker Heroes

Questions

Before Starting

Have you played this kind of game before?

“No”.

If not:

1. What do you expect this game to be about?
“Killing monsters by clicking them”.
2. What is your opinion of this genre?
“It’s good for kids, but not really in my demographic”.

During Play

What made you do this action (What were you thinking)?

“I clicked the monster to kill it”.

How did you know to do [action] to achieve [result]?

“Because the game is called “clicker heroes”, so I figured I should click”.

What made you do this action (What were you thinking)?

“I swept my mouse across the gold to collect it”.

How did you know to do [action] to achieve [result]?

“Because if you don’t collect the gold, you don’t get to keep it” (not true - potential UI issue).

What is grabbing most of your attention?

“The money and the monsters”.

What stands out as the most confusing?

“Nothing in particular, the interface is pretty easy to understand” (subject missed critical UI elements such as the secondary upgrades menu, but was apparently unaware of this).

Is there anything about the interface that annoys you?

“Some of the monster artwork annoys me”.

After Play

(If this is the first game like this that they have played) - Has your opinion of the genre changed at all, or do you still feel the same way about it?

“I still feel like it’s not very entertaining”.

Would you return to this website to play again? Why?

“Probably not, it’s sorta boring”.

What did you like the most?

“The rat sound effect”.

What did you like the least?

“The game didn’t have much depth”.

How intuitive was the interface? Were you confused by anything in particular?

“There needs to be a tutorial for the first level showing how to interact with the upgrades correctly”.

Would you play a game like this if it was a mobile app? What about if it was a windows executable? (If the game isn’t on a website, ask about that too).

“No, I probably wouldn’t play this game regardless of what platform it is on”.

If the cookies were replaced with something else, would it influence your enjoyment?

“No, anything is fine”.

If the game had a steampunk theme, what features would you expect? Any cool ideas?

“Replace the enemies with robots. Use a muted color style”.

Video: [Interview 3.mp4](#)

Participant 4

Game: Realm Grinder

Questions

Before Starting

Have you played this kind of game before?

If so:

1. What influences which games in this genre you choose to play?
Complexity.
This is the only clicker game that he plays because he likes the intricate class and currency system. Different classes have different advantages and disadvantages, which makes the game more fun.
2. What mood are you generally in when you play games like these?
Plays games like this when without internet or when tired. Finds the game to be relaxing.
3. How do games like this achieve mechanical depth? What makes them interesting?
Classes - different classes interact with different buildings in different ways.
As you progress, hidden features are unlocked via the artifact system.
Lots of play-style diversity, although it is good at easing the user into this complexity.
4. Do you prefer random luck-based progression or steady time-based progression?
Steady, time based progression.

If not:

1. What do you expect this game to be about?
2. What is your opinion of this genre?

During Play

What made you do this action (What were you thinking)?

How did you know to do [action] to achieve [result]?

What is grabbing most of your attention?

Currency, spell cooldowns, excavation costs.

What stands out as the most confusing? (*I think we should ask both at the end as an overall what was the most confusing thing, but also as they play, what is confusing or frustrating right now).

The Autocasting system

Automatic Casting

Cost: 100 B Coins

Allows autocasting of spells. To activate it, use the buttons in the spell's tooltip or by Ctrl+Clicking them. If you set a spell in Autocast mode, it will be fired automatically as soon as you have enough mana. If you have multiple spells in Autocast mode, the cheapest will be cast first. Later in the game, you will unlock more Autocasting options which will allow you to better manage your spells while idle.

While offline, spells are not cast. Instead, increase your mana produced amount by 10% of your Mana Regeneration per second.

Is there anything about the interface that annoys you?

Organization of upgrades. The upgrade menu is not organized well enough.

After Play

(If this is the first game like this that they have played) - Has your opinion of the genre changed at all, or do you still feel the same way about it?

Would you return to this website to play again? Why?

Doesn't play the game very often, but when they first started they played for hours every day for 2 weeks straight.

Are there any gameplay features that you wish were in the game?

No, 100% satisfied. The gameplay is perfectly satisfying as-is.

If you could change one thing about the interface, what would it be?

Options menu is clunky. Add tabs.

What did you like the most?

Excavation system - between reincarnations there is permanent progress. Also non-linear progression and traits.

What did you like the least?

The music is annoying. Games in this genre should not have music because it is designed to be played while doing something else.

How intuitive was the interface? Were you confused by anything in particular?

Slightly user unfriendly. 8.5/10 for usability.

Would you play a game like this if it was a mobile app? What about if it was a windows executable? (If the game isn't on a website, ask about that too).

Would not play on mobile because the UI would be too cramped.

If the cookies were replaced with something else, would it influence your enjoyment?

No, as long as the currency fits with the theme it is fine.

If the game had a steampunk theme, what features would you expect? Any cool ideas?

Buildings and occupations.

Additional Questions

a. How does the participant currently plan his/her activities?

Determines which upgrades fit best with current playstyle and builds up a “wishlist”. Then, buys these upgrades as they become available.

b. How does the participant currently monitor his/her activities?

c. How does the participant make sense of, or analyze his/her activities?

d. Does the participant share his/her activities/progress with others? How?

Yes. Would physically show the screen to others to share progress.

e. How does the participant stay motivated to do the activities, both when doing the activities and outside of the activities?

When in game - seeing progression and numbers go up.

When out of game - boredom.

f. Does the participant use any technologies or other resources (including people) to assist him/her in any of the above? Which ones? How are they used

Has used autoclickers, but feel that they reduce entertainment.

Has taught other people about the user interface setup - what production means.

Video: [Interview 4.mp4](#)

Participant 5

Game: Cookie Clicker

Video: [zoom_0.mp4](#)

Participant 6

Game: Cookie Clicker

Questions

Before Starting

Have you played this kind of game before?

If so:

1. What influences which games in this genre you choose to play?
2. What mood are you generally in when you play games like these?
3. How do games like this achieve mechanical depth? What makes them interesting?
4. Do you prefer random luck-based progression or steady time-based progression?

If not:

5. What do you expect this game to be about?

“It displays a cookie and as you click the numbers increase?”

6. What is your opinion of this genre?

“Not for me.”

During Play

What made you do this action (What were you thinking)?

“It had a big picture of a cookie on the side.”

How did you know to do [action] to achieve [result]?

“Every time I clicked on the cookie there was a plus one. I’m assuming that more numbers means... good.”

What is grabbing most of your attention?

“Just the cookie and how shiny it is.”

What stands out as the most confusing?

“I still don’t really get what the point of it is.”

Is there anything about the interface that annoys you?

“There aren’t really clear instructions.”

After Play

(If this is the first game like this that they have played) - Has your opinion of the genre changed at all, or do you still feel the same way about it?

“I still feel the same way about it.”

Would you return to this website to play again? Why?

“I would if I were paid to do it. Otherwise I would not because it’s not how I would like to spend my time.”

Are there any gameplay features that you wish were in the game?

“More instructions.”

If you could change one thing about the interface, what would it be?

“I don’t understand what the big space in the middle is for.”

What did you like the most?

“The picture of the cookie.”

What did you like the least?

“Lack of instructions.”

How intuitive was the interface? Were you confused by anything in particular?

“Yes, the three buttons on the right.”

Would you play a game like this if it was a mobile app? What about if it was a windows executable?

“If it was a mobile app.... yeah sure. If it was a windows executable, definitely not.”

If the cookies were replaced with something else, would it influence your enjoyment?

“Yes.”

If the game had a steampunk theme, what features would you expect? Any cool ideas?

“I don’t know what that is.”